

STT 465: Bayesian Data Analysis

Homework 2: Single-Parameter Bayesian Models

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Total: 5 points Due: October 9, 2026 at 5:00 PM

Instructions.

Show your derivations and justify your answers. You may use R for numerical calculations, simulation, and plotting, but you should first write the corresponding Bayesian model or probability expression. When R is requested, include both the relevant code and the numerical result.

Question 1: Bernoulli-Beta Posterior

[1 point]

A basketball player makes 18 of 25 free throws. Let

$$X_i \mid \omega \sim \text{Bern}(\omega), \quad i = 1, \dots, 25,$$

where ω is the player's underlying probability of making a free throw.

Assume the prior

$$\omega \sim \text{Beta}(2, 2).$$

- (a) Write the likelihood function

$$L(\omega \mid \mathbf{x})$$

up to a multiplicative constant that does not depend on ω .

[0.2 pt]

- (b) Use Bayes' theorem to derive the posterior distribution

$$\pi(\omega \mid \mathbf{x}).$$

Identify the posterior distribution and its parameters.

[0.2 pt]

- (c) Derive the posterior mean

$$E[\omega \mid \mathbf{x}]$$

and calculate its numerical value.

[0.2 pt]

- (d) Calculate the posterior mode and posterior median. You may use `qbeta()` for the median.

[0.2 pt]

- (e) Use R to calculate

$$P(\omega > 0.70 \mid \mathbf{x}).$$

Include the R command you use.

[0.2 pt]

Question 2: Prior Sensitivity and Bayesian Updating [1 point]

Suppose a treatment is effective for 18 of 21 patients. Let ω denote the probability that the treatment is effective.

Consider the following two prior distributions:

$$\text{Prior A: } \omega \sim \text{Beta}(1, 1),$$

and

$$\text{Prior B: } \omega \sim \text{Beta}(4, 0.9).$$

- (a) Derive the posterior distribution under Prior A.

[0.2 pt]

- (b) Derive the posterior distribution under Prior B.

[0.2 pt]

- (c) For a general prior

$$\omega \sim \text{Beta}(a, b),$$

show that the posterior mean can be written as

$$E[\omega \mid \mathbf{x}] = \frac{a+b}{a+b+n} \frac{a}{a+b} + \frac{n}{a+b+n} \frac{x}{n}.$$

Explain what the two terms represent.

[0.2 pt]

- (d) Compute the posterior mean under Priors A and B. Briefly explain why the two answers differ.

[0.2 pt]

- (e) Use R to plot the two posterior densities on the same graph. Add a vertical line at the sample proportion

$$\hat{\omega} = \frac{18}{21}.$$

Briefly describe what the plot shows.

[0.2 pt]

Question 3: Credible Intervals and HPDI

[1 point]

Suppose that after observing data, the posterior distribution is

$$\omega \mid \mathbf{x} \sim \text{Beta}(22, 4).$$

- (a) Write the R commands needed to calculate the 95% equal-tailed credible interval

$$\left[F_{\omega \mid \mathbf{x}}^{-1}(0.025), F_{\omega \mid \mathbf{x}}^{-1}(0.975) \right].$$

Report the numerical interval.

[0.2 pt]

- (b) Explain in one or two sentences the Bayesian interpretation of this 95% credible interval.

[0.2 pt]

- (c) Generate 10,000 posterior samples using

```
set.seed(1)
omega_post <- rbeta(10000, shape1 = 22, shape2 = 4)
```

Use the `HDInterval` package to estimate the 95% HPDI.

[0.2 pt]

- (d) Compare the equal-tailed credible interval and the HPDI. Which one is shorter?

[0.2 pt]

- (e) Explain why the two intervals need not be identical for a skewed posterior distribution.

[0.2 pt]

Question 4: Comparing Two Population Proportions

[1 point]

A study of births reports that among 980 births in one group, 437 were female. In a second group, among 820 births, 398 were female.

Let

$$\omega_x = \text{female birth proportion in group 1,}$$

and

$$\omega_y = \text{female birth proportion in group 2.}$$

Assume independent priors

$$\omega_x \sim \text{Beta}(1, 1), \quad \omega_y \sim \text{Beta}(1, 1).$$

- (a) Derive the posterior distributions of ω_x and ω_y .

[0.2 pt]

- (b) Calculate the posterior means

$$E[\omega_x \mid \mathbf{x}] \quad \text{and} \quad E[\omega_y \mid \mathbf{y}].$$

[0.2 pt]

- (c) Use the following posterior simulation idea to estimate

$$P(\omega_x < \omega_y \mid \mathbf{x}, \mathbf{y}).$$

Complete the missing parts of the R code.

```
set.seed(1)
U <- 10000

omega_x <- rbeta(
  U,
  shape1 = _____,
  shape2 = _____
)

omega_y <- rbeta(
  U,
  shape1 = _____,
  shape2 = _____
)

mean(omega_x < omega_y)
```

[0.2 pt]

- (d) Explain in words what the probability

$$P(\omega_x < \omega_y \mid \mathbf{x}, \mathbf{y})$$

means.

[0.2 pt]

- (e) A frequentist analysis uses

```
prop.test(
  c(437, 398),
  c(980, 820),
  alternative = "less"
)
```

Briefly explain the difference between the posterior probability in part (c) and the frequentist p -value.

[0.2 pt]

Question 5: Poisson-Gamma Model and Prediction [1 point]

Suppose the number of customer complaints received by a company on day i satisfies

$$X_i \mid \lambda \sim \text{Poi}(\lambda), \quad i = 1, \dots, n.$$

Over 20 days, a total of 54 complaints are observed:

$$n = 20, \quad \sum_{i=1}^{20} x_i = 54.$$

Assume the Gamma prior

$$\lambda \sim \text{Ga}(a = 2, b = 1),$$

where a is the shape parameter and b is the rate parameter.

(a) Show that the likelihood function has the form

$$L(\lambda \mid \mathbf{x}) \propto \lambda^{\sum_i x_i} \exp(-n\lambda).$$

[0.2 pt]

(b) Derive the posterior distribution of λ and identify its parameters.

[0.2 pt]

(c) Calculate the posterior mean

$$\text{E}[\lambda \mid \mathbf{x}]$$

and a 95% equal-tailed credible interval using `qgamma()`.

[0.2 pt]

(d) Derive the posterior predictive mean for the number of complaints on a future day,

$$\text{E}[X_f \mid \mathbf{x}].$$

Explain why it is equal to

$$\text{E}[\lambda \mid \mathbf{x}].$$

[0.2 pt]

(e) Use posterior simulation to estimate

$$P(X_f \geq 4 \mid \mathbf{x}).$$

You may use the following structure:

```
set.seed(1)
U <- 10000

lambda_post <- rgamma(
  U,
  shape = _____,
  rate = _____
)

x_future <- rpois(U, lambda = lambda_post)

mean(x_future >= 4)
```

Report your estimate and explain what it means.

[0.2 pt]